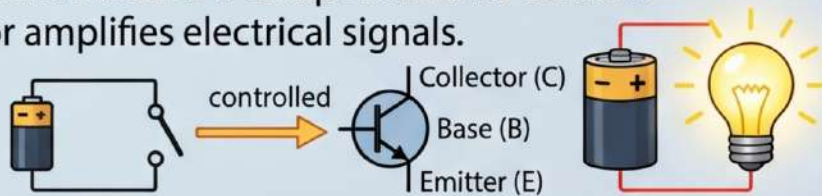


THE TRANSISTOR: Explained 10x Simpler

🔌 What is a Transistor?

A transistor is a component that controls or amplifies electrical signals.

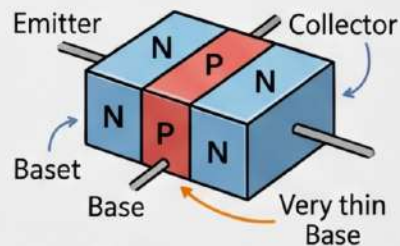


Essentially, it acts as a **digital switch** (turning ON or OFF current flow) or an **analog amplifier** (boosting signal strength).

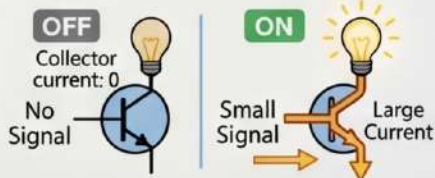
🧱 How is a Transistor Made?

➡ **Very simple:**

- 3 semiconductor layers
- 3 terminals (Base, Collector, Emitter)



Controlling and Amplifying



This dual capability makes transistors the fundamental building block of all modern digital and analog electronics.

A small input signal to the 'Base' controls and opens a much larger current flow between 'Collector' and 'Emitter'.